

Predicting Wildfire Threat to Evacuation Zones Using Survival Analysis

WiDS Global Datathon 2026 | Bruna Macedo Porto | MS Data Science, Seattle University

Kaggle Leaderboard
Score: 0.9663

THE PROBLEM

Task: Predict P(fire threatens evacuation zone) at 12h, 24h, 48h, 72h using only the first 5 hours of data.

Dataset: 221 wildfire events, 34 features, 69% censored.

Metric: $0.3 \times \text{C-index} + 0.7 \times (1 - \text{Weighted Brier})$. Calibration is 70% of the score.

KEY EDA FINDINGS

Perfect 5km threshold

Every fire within 5km hit (100%, n=69). Every fire beyond 5km did not (0%, n=152).

Most hits happen fast

71% within 12h, 91% within 24h, 96% within 48h.

89% feature sparsity

Fires with growth hit at 71% vs 26% without.

Distance × alignment interaction

Close + aligned → 76% hit. Far → 0% regardless.

Log-rank tests confirm significance

Distance, growth, alignment, movement all $p < 0.001$. Temporal flags not significant ($p > 0.2$).

FEATURE ENGINEERING

16 features engineered from 34 raw inputs

Binary indicators (5): has_growth, has_movement, has_alignment, etc.

Distance transforms (2): log(distance), distance in km

Interactions (3): alignment/distance, closing/distance, growth/distance

5 features achieve 97% of full performance.

MODEL COMPARISON

Model	CV Hybrid	CV Brier	LB Score
Cox PH	0.948	0.042	—
Random Survival Forest	0.964	0.019	0.958
Gradient Boosting Surv.	0.974	0.016	0.963
GBS Risk + Platt Cal.	0.974	0.015	0.964
82% Risk+Platt + 18% Cox	0.974	0.016	0.9663

Shallow trees dominate: max_depth=2 beats deeper models on 221 rows.

THE WINNING APPROACH

1. **GBS for ranking** — risk scores rank fires by urgency

2. **Platt scaling for calibration** — 2 parameters per horizon, prevents overfitting

3. **Cox PH for diversity** — different errors improve generalization

82% Risk+Platt + 18% Cox → LB 0.9663

KEY INSIGHTS

The ensemble paradox

Adding Cox (weakest model) to GBS (strongest) improves test performance. Diversity beats optimization.

Calibration overfitting

Isotonic (n−1 params) hurt the LB by 0.01. Platt (2 params) helped. Parametric beats non-parametric on small data.

CV-LB divergence is real

Best CV model (0.976) didn’t produce the best LB score. Ensembles that score lower in CV generalize better.

CHALLENGES

Small dataset (221 rows) - every modeling decision risks overfitting

Calibration overfitting - isotonic calibration improved in-sample Brier but dropped the LB score

69% censoring rate - most fires never hit during the observation window

CROSS-VALIDATION

5-Fold Stratified: Hybrid 0.972

Leave-One-Out: Hybrid 0.971

Close agreement confirms robustness.

NEXT STEPS

Beta calibration

BART-Survival

Mixture cure models